LinkedIn edunet foundation

AI CAREERS FOR WOMEN
LINKEDIN FELLOWSHIP

Sales Performance & Demand Analysis

Project Domain

Excel / Power BI

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business Intelligence

by:

Faculty Name
Sagar Chetan Mehta

Student Email
nadkaramruta42@gmail.com

Guide
Umamaheswari

Institution:

Smt. Maniben M.P. Shah Women's College of Arts & Commerce

Year - 2026

3. Problem Statement

In today's competitive market, businesses generate large volumes of sales data but often fail to utilize it effectively for decision-making. One of the major challenges faced by organizations is the **lack of accurate demand forecasting and clear visibility into sales performance**, which directly impacts profitability and operational efficiency.

Companies frequently encounter:

- **Overstocking or stock shortages** due to poor demand planning
- **Limited visibility into product and regional performance**
- **Difficulty in identifying profitable products and trends**
- **Delayed decision-making due to lack of interactive reporting systems**

These issues lead to **revenue loss, increased operational costs, and inefficient business strategies**.

This project aims to solve these challenges by using **Excel and Power BI** to analyze sales data, predict demand trends, and generate meaningful business insights. The goal is to enable organizations to make **data-driven decisions**, improve inventory planning, and enhance overall sales performance.

4. Objectives

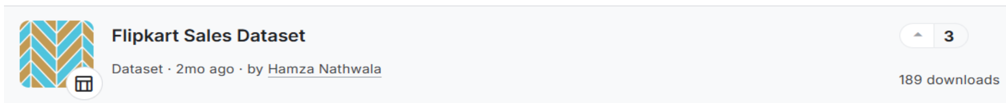
List the specific goals your project aims to accomplish. Objectives should be clear and measurable.

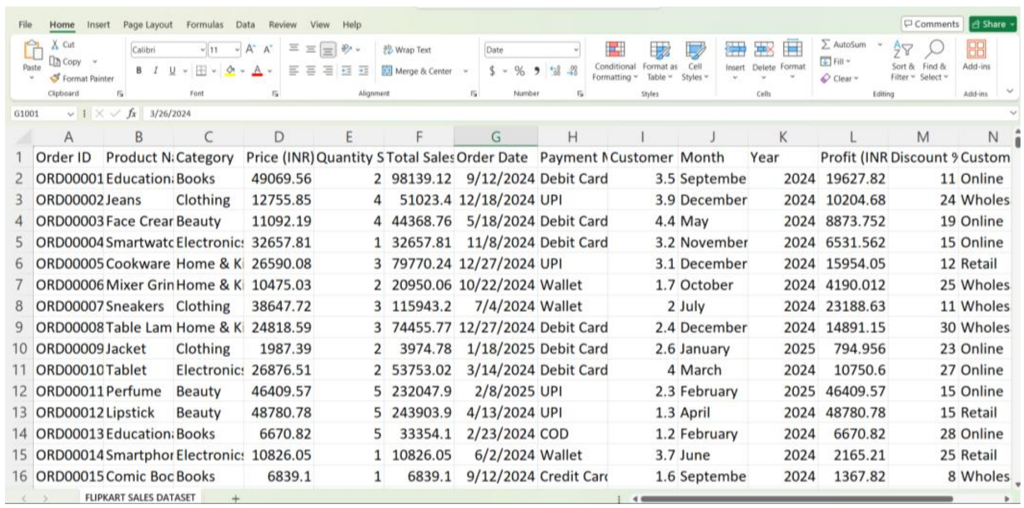
- To explore and understand the structure of the Flipkart sales dataset including product category, sales amount, profit, region, and customer segment.
- To clean and preprocess the dataset by removing duplicates, handling missing values, and correcting data formats.
- To analyze sales performance based on **product category, region, and customer segment**.
- To identify **top-selling products, highest revenue categories, and profitable regions**.
- To create an **interactive dashboard in Excel** showing key sales KPIs.
- To use **AI tools such as ChatGPT** to generate insights and summaries from the analyzed data.

5. Methodology / Implementation Plan

Provide a simple overview of how the project will be carried out. The following five-phase workflow applies to all projects under this fellowship program.

#	Phase	Description
1	**Data Collection & Exploration**	The project uses a Flipkart sales dataset containing information such as product name, category, price, quantity sold, order date, customer rating, region, and profit. The dataset will be explored to understand field names, data types, and data distribution.

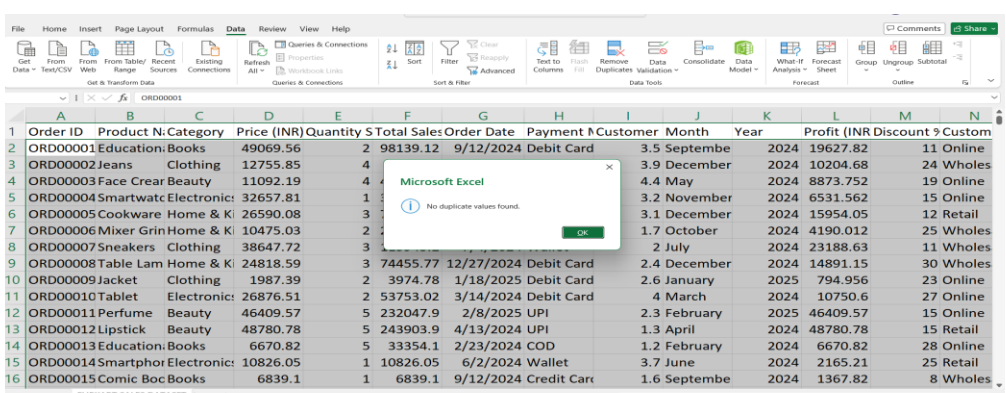




	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Order ID	Product N	Category	Price (INR)	Quantity S	Total Sales	Order Date	Payment M	Customer	Month	Year	Profit (INR	Discount %	Custom
2	ORD00001	Education	Books	49069.56	2	98139.12	9/12/2024	Debit Card	3.5	Septembe	2024	19627.82	11	Online
3	ORD00002	Jeans	Clothing	12755.85	4	51023.4	12/18/2024	UPI	3.9	December	2024	10204.68	24	Wholes
4	ORD00003	Face Crear	Beauty	11092.19	4	44368.76	5/18/2024	Debit Card	4.4	May	2024	8873.752	19	Online
5	ORD00004	Smartwatc	Electronic	32657.81	1	32657.81	11/8/2024	Debit Card	3.2	November	2024	6531.562	15	Online
6	ORD00005	Cookware	Home & Ki	26590.08	3	79770.24	12/27/2024	UPI	3.1	December	2024	15954.05	12	Retail
7	ORD00006	Mixer Grin	Home & Ki	10475.03	2	20950.06	10/22/2024	Wallet	1.7	October	2024	4190.012	25	Wholes
8	ORD00007	Sneakers	Clothing	38647.72	3	115943.2	7/4/2024	Wallet	2	July	2024	23188.63	11	Wholes
9	ORD00008	Table Lam	Home & Ki	24818.59	3	74455.77	12/27/2024	Debit Card	2.4	December	2024	14891.15	30	Wholes
10	ORD00009	Jacket	Clothing	1987.39	2	3974.78	1/18/2025	Debit Card	2.6	January	2025	794.956	23	Online
11	ORD00010	Tablet	Electronic	26876.51	2	53753.02	3/14/2024	Debit Card	4	March	2024	10750.6	27	Online
12	ORD00011	Perfume	Beauty	46409.57	5	232047.9	2/8/2025	UPI	2.3	February	2025	46409.57	15	Online
13	ORD00012	Lipstick	Beauty	48780.78	5	243903.9	4/13/2024	UPI	1.3	April	2024	48780.78	15	Retail
14	ORD00013	Education	Books	6670.82	5	33354.1	2/23/2024	COD	1.2	February	2024	6670.82	28	Online
15	ORD00014	Smartphor	Electronic	10826.05	1	10826.05	6/2/2024	Wallet	3.7	June	2024	2165.21	25	Retail
16	ORD00015	Comic Boc	Books	6839.1	1	6839.1	9/12/2024	Credit Carc	1.6	Septembe	2024	1367.82	8	Wholes

FLIPKART SALES DATASET

#	Phase	Description
2	**Data Cleaning & Preprocessing**	The dataset will be cleaned by removing duplicate records, handling missing values, correcting date and numerical formats, and standardizing text fields such as product categories and regions.



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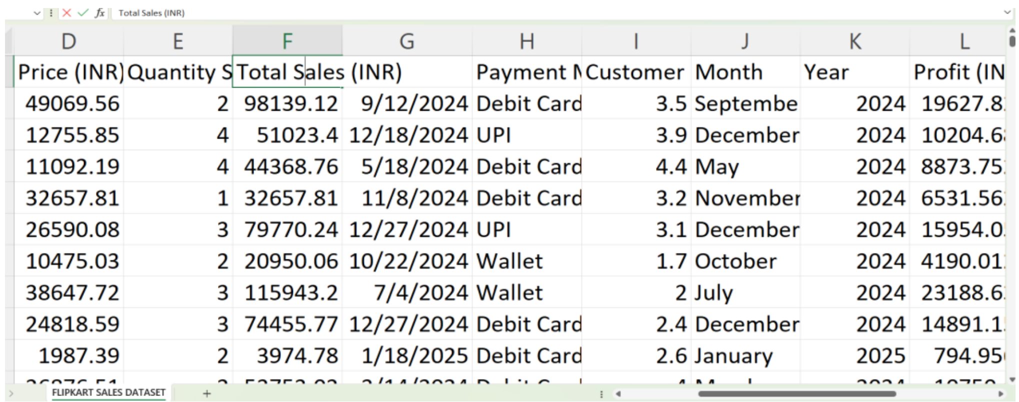
<table>
<tr>
<td>3</td>
<td>Analysis & AI Integration</td>
<td>Excel formulas, pivot tables measures will be used to analyze key metrics such as total sales, profit, quantity sold, and average customer ratings. AI tools will be used to generate summaries and insights from the analysis.</td>
</tr>
<tr>
<td>4</td>
<td>Visualization / Output</td>
<td>An interactive dashboard will be created showing:

• Total Sales and Profit KPIs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<tr><th>D</th><th>E</th><th>F</th><th>G</th><th>H</th><th>I</th><th>J</th><th>K</th><th>L</th></tr>
<tr><td>Price (INR)</td><td>Quantity S</td><td>Total Sales (INR)</td><td></td><td>Payment N</td><td>Customer</td><td>Month</td><td>Year</td><td>Profit (IN</td></tr>
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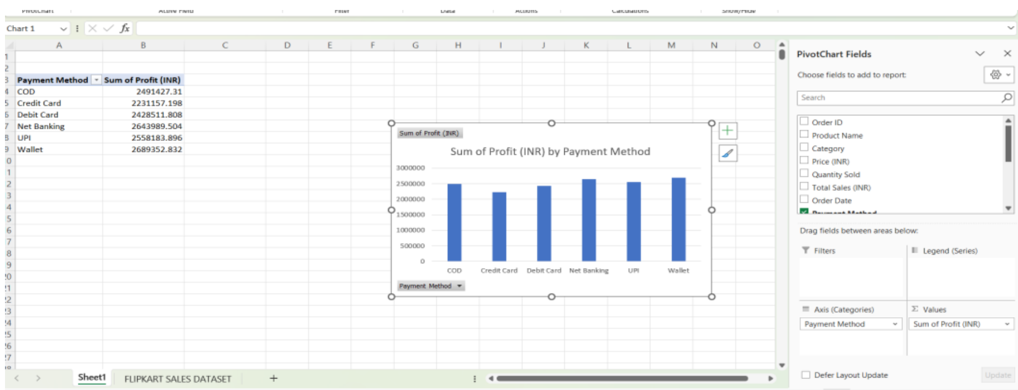
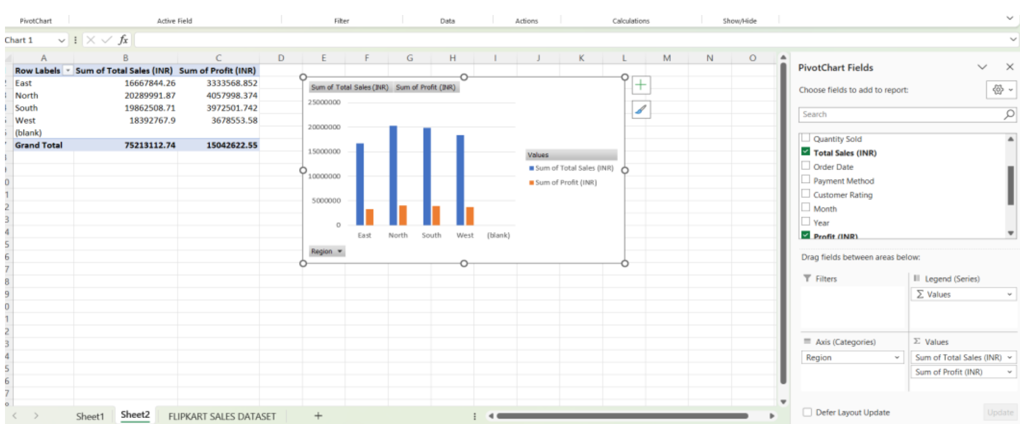
• Category-wise Sales



• Region-wise Sales Performance



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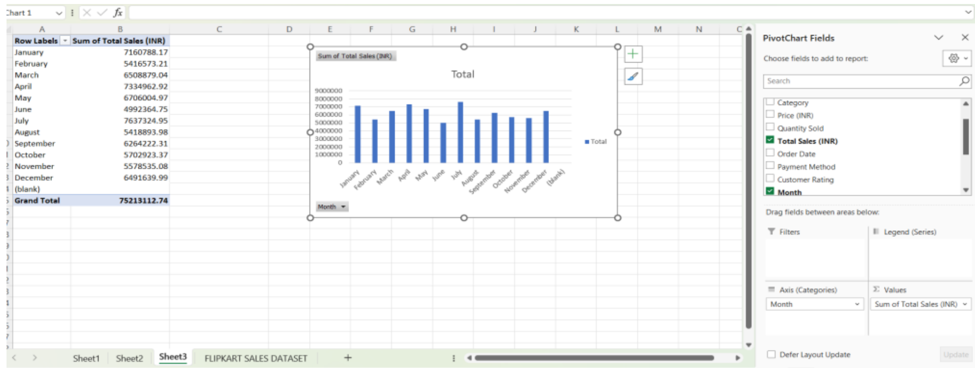
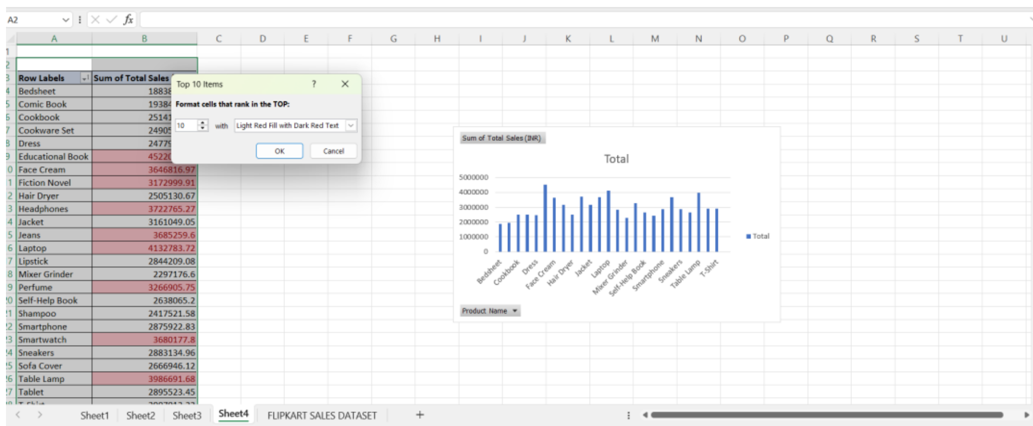
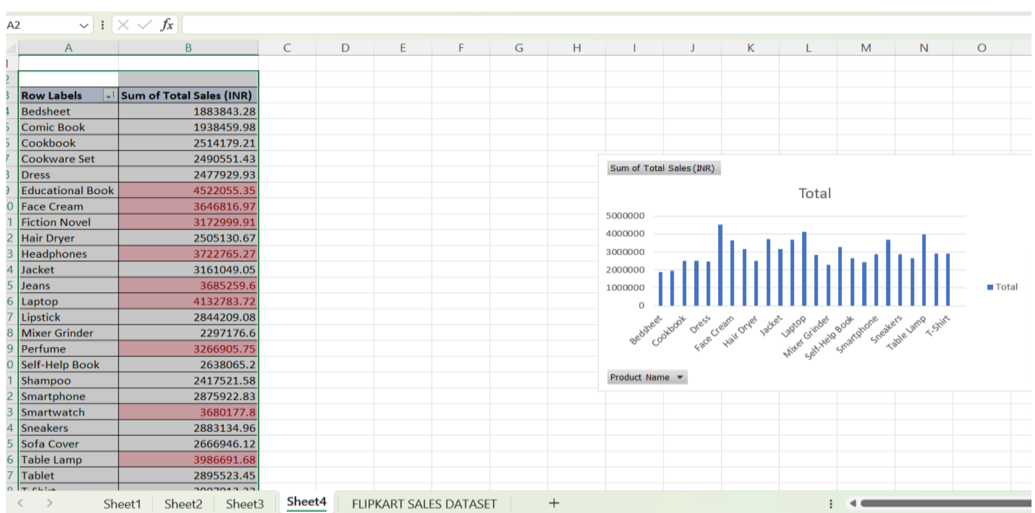
- Monthly Sales Trends



Row Labels	Sum of Total Sales (INR)
January	7160788.17
February	5416573.21
March	6508879.04
April	7334962.92
May	6706004.97
June	4992364.75
July	7637324.95
August	5418893.98
September	6264222.31
October	5702923.37
November	5578535.08
December	6491639.99
(blank)	
Grand Total	75213112.74

- Top Performing Products





Row Labels	Sum of Total Sales (INR)
Bedsheet	1883843.28
Comic Book	1938459.98
Cookbook	2514179.21
Cookware Set	2490551.43
Dress	2477929.93
Educational Book	4522055.35
Face Cream	3646816.97
Fiction Novel	3172999.91
Hair Dryer	2505130.67
Headphones	3722765.27
Jacket	3161049.05
Jeans	3685259.6
Laptop	4132783.72
Lipstick	2844209.08
Mixer Grinder	2297176.6
Perfume	3266905.75
Self-Help Book	2638065.2
Shampoo	2417521.58
Smartphone	2875922.83
Smartwatch	3680177.8
Sneakers	2883134.96
Sofa Cover	2666946.12
Table Lamp	3986691.68
Tablet	2895523.45

5	**Documentation & Submission**	The final report will document the complete workflow including dataset cleaning, analysis, dashboard creation, and AI-assisted insights.

6. Resources and Timeline

A. Hardware & Software Requirements

- Laptop or PC — Intel Core i3 or higher, minimum 4 GB RAM (8 GB recommended)
- Operating System: Windows 10 / 11 or macOS
- Microsoft Excel (Microsoft 365 / Excel 2019 or later)
- Power BI Desktop — latest version (free download from Microsoft) — where applicable
- Internet connection — required for Copilot and AI Chatbot access
- Web browser — Google Chrome or Microsoft Edge (latest version)

B. Dataset

- **Dataset Availability**
  The dataset used for this project is already available in CSV format and contains sales-related information such as product name, category, price, quantity sold, discount, region, profit, and customer segment. This dataset represents e-commerce sales transactions that can be analyzed to understand business performance.

- **Dataset Source**
  The dataset is publicly available from **Kaggle**, a popular platform that provides open datasets for data analysis and machine learning projects. It is a sample sales dataset inspired by the operations of **Flipkart**, used for educational and analytical purposes.

- **Data Usage in the Project**
  The dataset will be imported into Microsoft Excel where it will be cleaned, processed, and analyzed. Techniques such as removing duplicates, handling missing values, and standardizing formats will be applied before performing analysis.

- **Purpose of Using the Dataset**
  The dataset will help analyze key business metrics such as total sales, profit, regional performance, product demand, and customer segments. These insights will support data-driven decision making and demonstrate practical data analytics skills.

C. Project Timeline

A simple table showing planned tasks and expected completion dates.

Phase	Task	Tentative Completion Date
1	Dataset collection, exploration, and understanding	*Week 1*
2	Data cleaning, preprocessing, and validation / Prompt design	*Week 2*

3	Analysis, AI tool integration (Copilot / Chatbot), and insight generation	*Week 2 – 3*
4	Dashboard / output build, testing, and review	*Week 3*
5	Report writing, GitHub upload, and video recording and submission	*Week 4*

7. Expected Outcomes / Conclusion

- Predict future sales demand using **Excel forecasting functions** to improve inventory planning and reduce stock issues.
- Analyze key performance indicators like **sales, profit, and growth** using Excel and visualize them through **Power BI dashboards**.
- Identify **top-performing and low-performing products and regions** to support better business decisions.
- Evaluate **product profitability** using Excel formulas and present insights using Power BI visualizations.
- Develop an **interactive dashboard** for easy monitoring of sales trends and business performance.
- Enable **data-driven decision-making**, helping businesses improve efficiency, profitability, and overall strategy.

This project demonstrates how **Excel and Power BI** can be effectively used to transform raw sales data into meaningful business insights. By focusing on demand prediction, sales performance, and profitability analysis, the project enables organizations to improve **inventory planning, strategic decision-making, and overall business growth**.

8. References

List all references, tutorials, and documentation used in the project. Students must cite any material they referred to, including research papers, GitHub repositories, YouTube tutorials, blogs, and official documentation.

- Edunet Foundation. AICW Fellowship Program Guidelines. 2026. Available at: https://www.edunetfoundation.org

- Microsoft Corporation. Microsoft Excel Documentation. Available at: https://support.microsoft.com/en-us/excel
- Microsoft Corporation. Power BI Documentation. Available at: https://learn.microsoft.com/en-us/power-bi/
- Kaggle Inc. Open Datasets. Available at: https://www.kaggle.com/datasets